# Deep Learning (23AID211)

## Bias and Variance

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# Bias and Variance in Neural Network Training

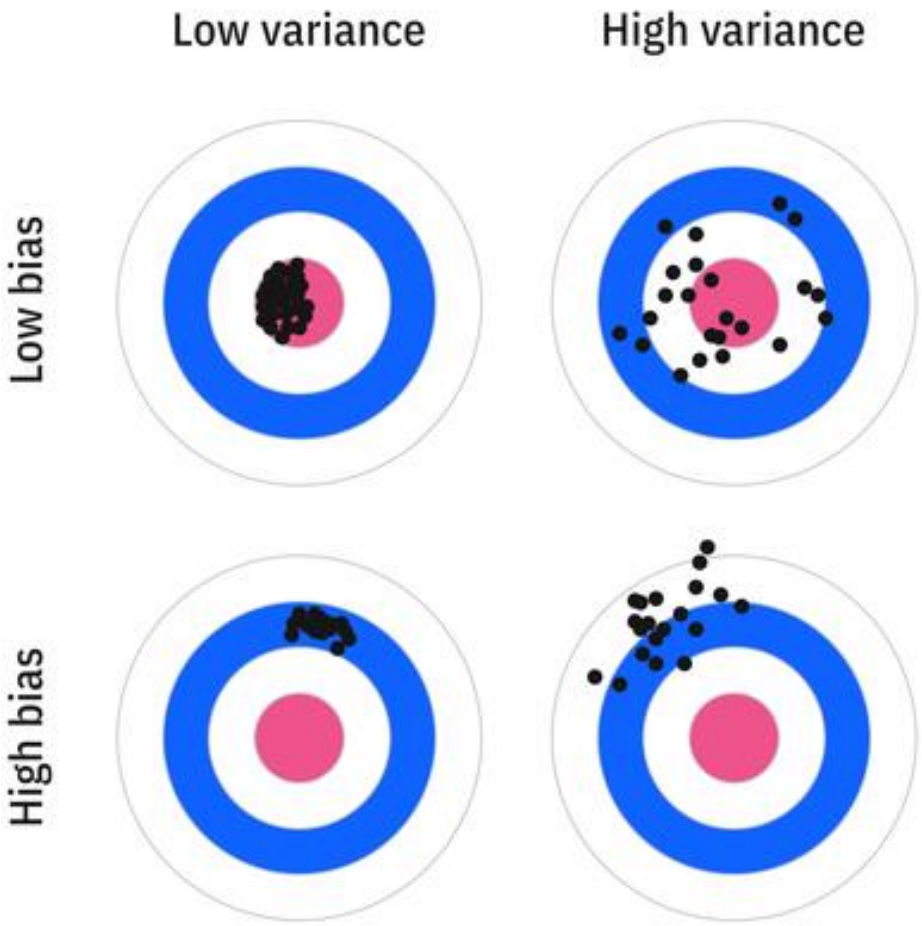
## Bias

- **Bias (systematic error)** is defined as the difference between average prediction and actual value.
- A model with **high bias** does not match well and leads to high errors on training data as well as test data.
- While the model with **low bias** matches the training data well (high training accuracy). It leads to low error in training data but high error in test data.
- **High Bias –** occurs due to erroneous assumptions in the machine learning model. Models with high bias cannot capture the hidden pattern in the training data. This leads to underfitting.
- **Low Bias –** can capture the hidden pattern in the training data. Low bias leads to high variance and, eventually, overfitting.

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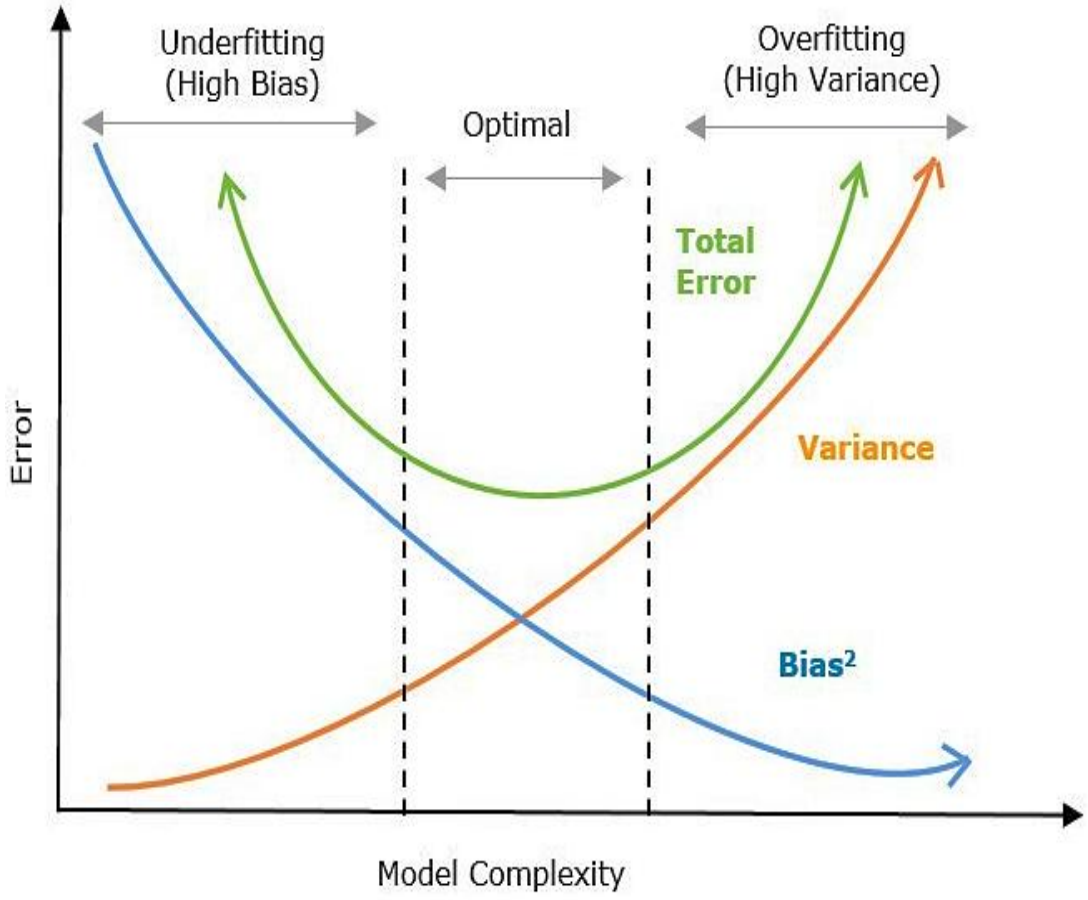
## Variance

- **Variance** measures how much a model's predictions change with different training datasets.
- **High-variance** models are sensitive to noise in the training data and cause **overfitting**. *A model with complex architecture and more parameters tends to have high variance and low bias*.
- **Low Variance**–model with is unable to capture the hidden pattern in the data. Low variance may occur when we have a very small amount of data or use a very simplified model. Low variance leads to **underfitting**.



# Bias and Variance Tradeoff

- The bias-variance tradeoff is finding a balance between the error introduced by bias and the error introduced by variance.
- With increased model complexity, the bias will decrease, but the variance will increase.
- However, when we decrease the model complexity, the bias will increase, and the variance will decrease. So we need a balance between bias and variance so total prediction error is minimized.
- When bias reduces, variance will increase. So we need to find an optimal bias and variance such that the prediction error is minimized.



# Bias-Variance Tradeoff

- They overfit the training set and fail to generalize to unseen data, leading to overfitting and leading to predictions with abnormally **high variability.**

- **High-bias models** are typically too simplistic to capture the true patterns in the data. They underfit the training set, leading to poor training and test accuracy.

The prediction error in the machine learning model can be written mathematically as follows: *Error = bias2 + variance + irreducible error*

# Techniques to Balance Bias and Variance

## Reducing High Bias

- **Choosing a more complex model** – choosing a more complex model may reduce the bias error of the model prediction.
- **Adding more features** – Adding mode features can increase the complexity of the model that can capture even better hidden patterns that will decrease the bias error of the model.
- **Reducing regularization** – it prevents overfitting, but while decreasing the variance, it can increase bias. So, reducing the regularization parameters or removing regularization overall can reduce bias errors.

## Reducing High Variance

- **Applying regularization techniques** – add penalty to complex model that will eventually result in reduced complexity of the model.
- **Simplifying model complexity** – A less complex model will have low variance. You can reduce the variance by using a simpler algorithm.
- **Adding more data –** Adding more data to the dataset can help the model to perform better showing less variance.
- **Cross-validation** – Cross-validation can be useful to identify overfitting by comparing the performance on training and validation sets of the datasets.